Lab exercise 4 --- Signals and Systems

Reference book:

John R.Buck, Michael M. Daniel, Andrew C. Singer. Computer Exploration in Signals and Systems —— Using MATLAB.

Objective:

1. Understand the sampling theorem, and verify the aliasing phenomeno due to undersampling.
2. Perform Single-Sideband AM with Hilbert Transform.
3. Make pole-zero plot for CT and DT system.
4. Understand the pole location's `influence` on the frequency response of a system.

Related commands:

fft, freqs, freqz, filter, conv, roots, zplane, dpzplot

MATLAB exercises:

1. Verify the aliasing phenomeno due to undersampling. 7.1 (a)(b)(c)(d)
2. Perform Single-Sideband AM with Hilbert Transform. 8.1 (c)(d)(e) (i)-(n)
3. Make pole-zero plot using MATLAB. 9.1 (a)(c) 10.1(a)(b)
4. Obtain the frequency response of a second-order system. 9.2 (a)(b)

1. Verify the aliasing phenomeno due to undersampling.

7.1 (a)(b)(c)(d)

a)b)

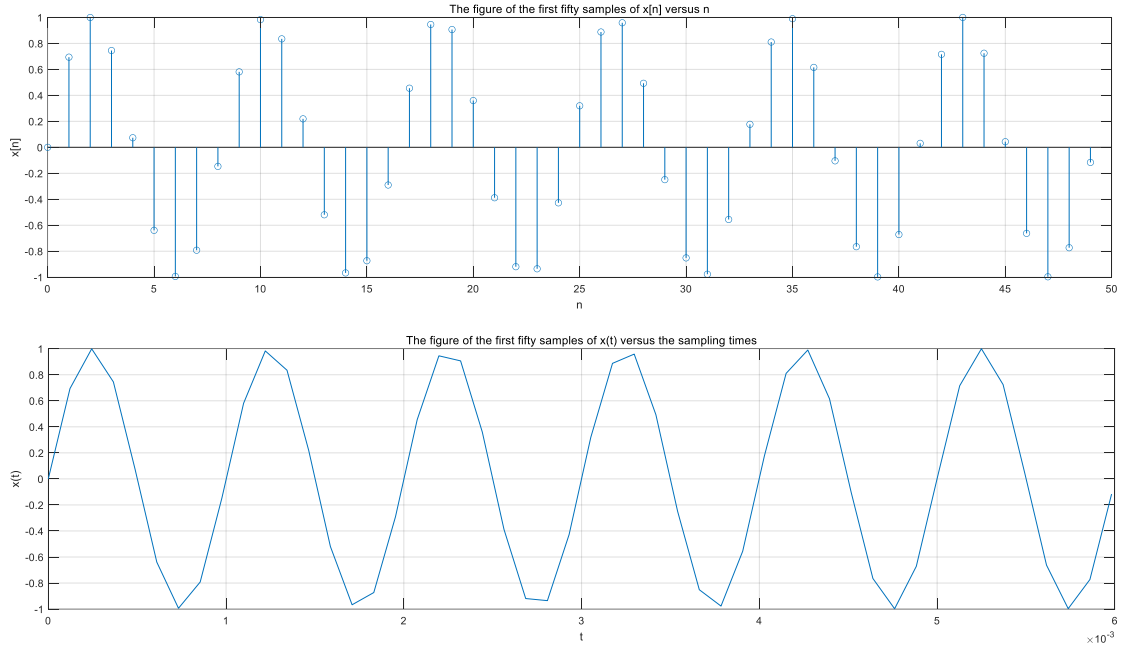
code:

```
clc
clear all
n=[0:8191];
T=1/8192;
t=n./8192;
x=sin(2000*pi*t);
figure;
subplot(2,1,1);
stem(n(1:50),x(1:50));
xlabel('n');
ylabel('x[n]'),
title('The figure of the first fifty samples of x[n] versus n');
grid on;
subplot(2,1,2);
plot(t(1:50),x(1:50));
xlabel('t');
ylabel('x(t)'),
title('The figure of the first fifty samples of x(t) versus the
```

sampling times');

grid on;

Output:



c)

code：

clc

clear all

n=[0:8191];

T=1/8192;

t=n./8192;

x=sin(2000*pi*t);

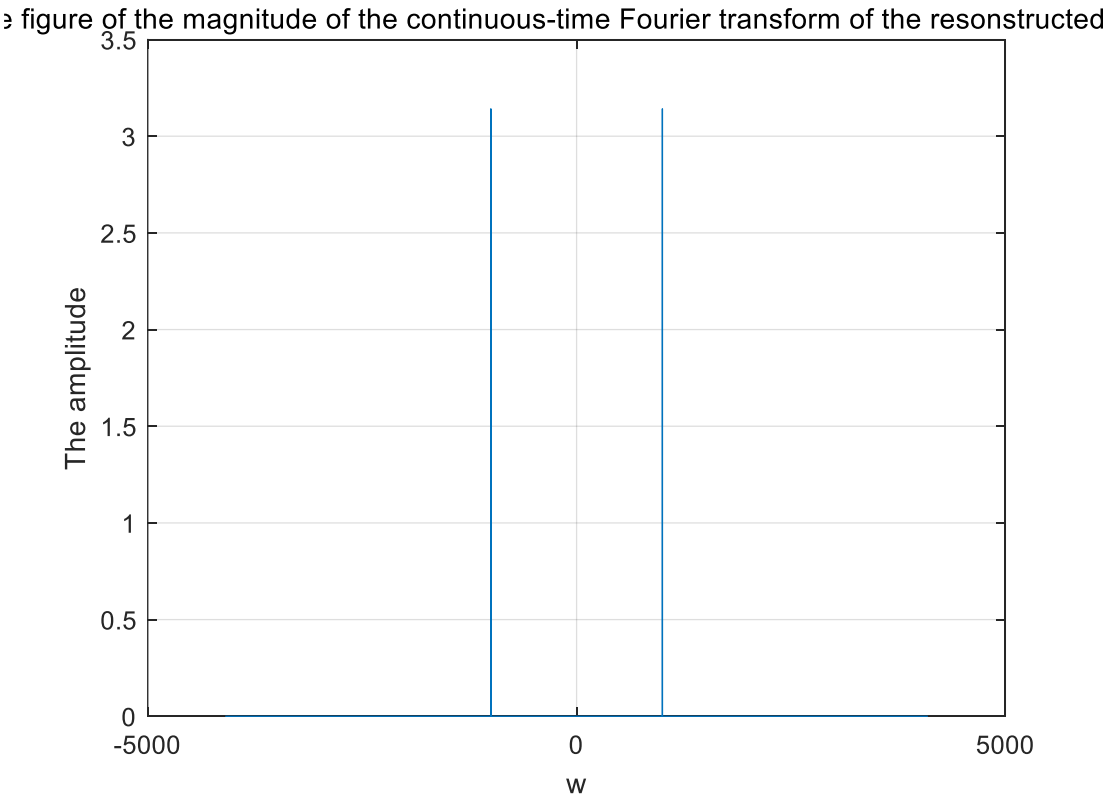
[X,w]=ctfts(x,T);

figure;

plot(w,abs(X));

```
xlabel('w');
ylabel('The amplitude');
title('The figure of the magnitude of the continuous-time Fourier transform of the resonstructed signal');
grid on;
% index=find(abs(X)>1);
% alpha=angle(X(index))*180/pi
```

Output:



Answer:

- X is nonzero at the proper frequency values ±1000 rad/s, which matches the expected frequencies of the input sine wave.

- Besides, the small nonzero values at other frequencies are caused by numerical round-off errors.

  The phase of is correct at the proper frequency values ±1000 rad/s, where ｜X｜is significant.
- At frequencies where ｜X｜is nearly zero, the phase can be assumed to be zero, as the small values are due to numerical round-off error.

  d)

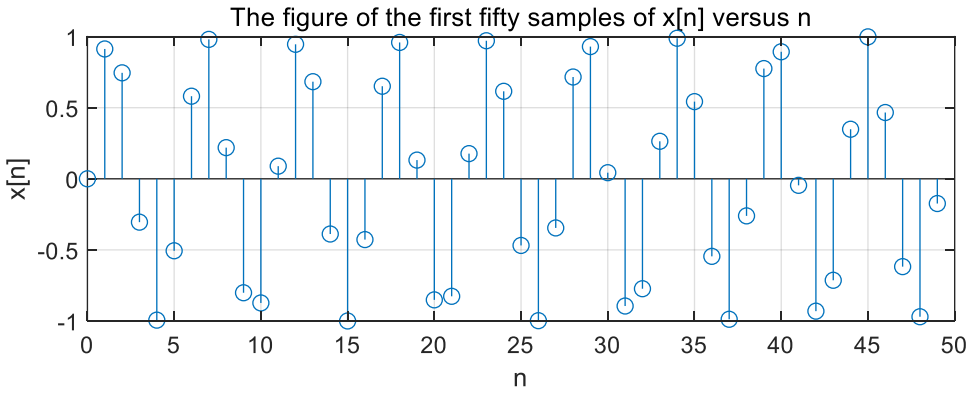
  code:

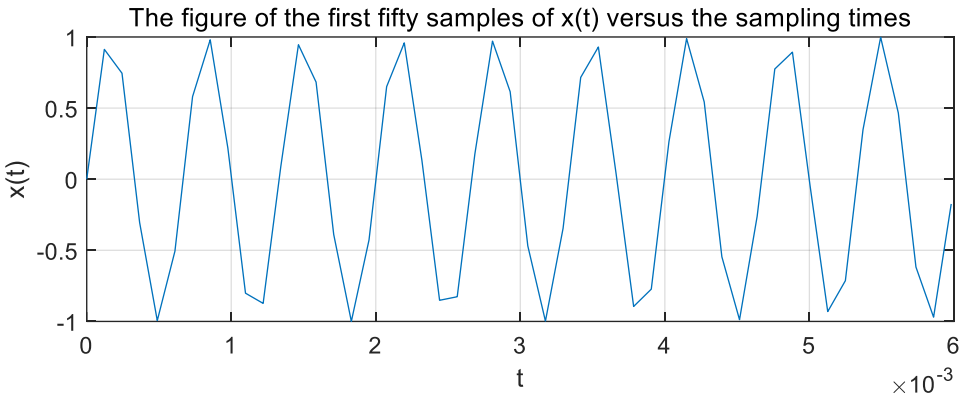
  ($\Omega = 2\pi(1500)$)

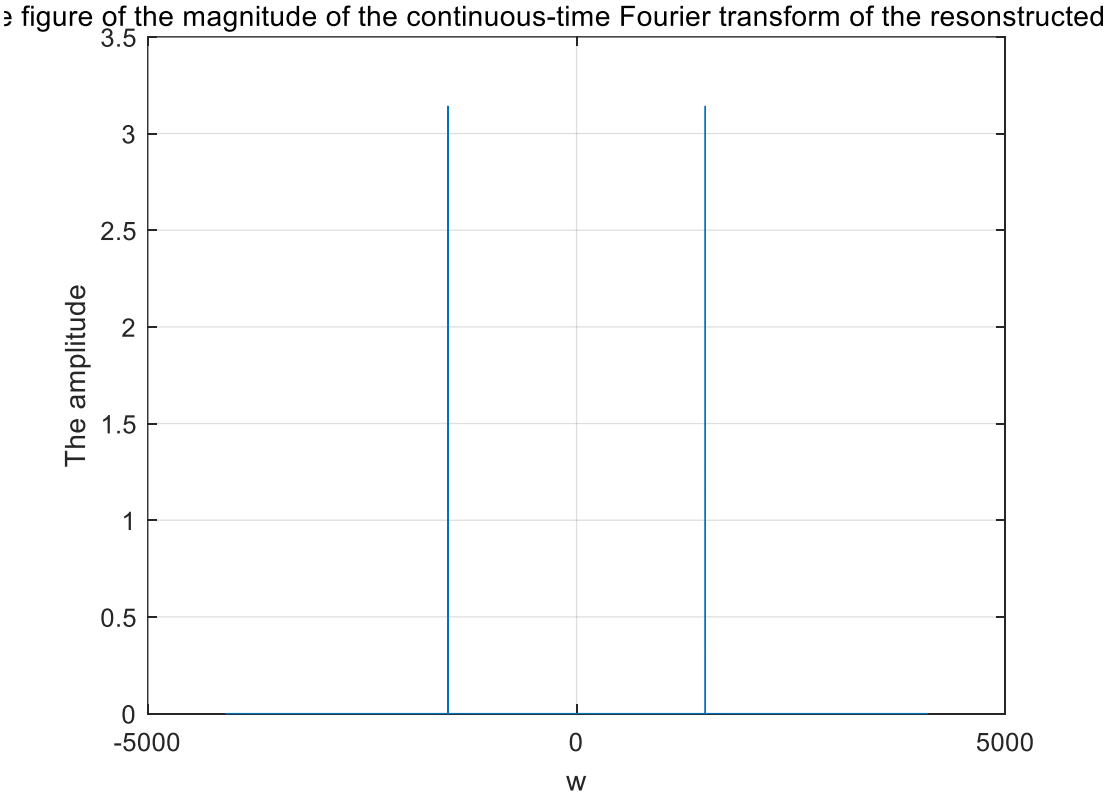
```
clc
clear all
n=[0:8191];
T=1/8192;
t=n./8192;
x=sin(3000*pi*t);
figure;
subplot(2,1,1);
stem(n(1:50),x(1:50));
xlabel('n');
ylabel('x[n]');
title('The figure of the first fifty samples of x[n] versus n');
```

```
grid on;
subplot(2,1,2);
plot(t(1:50),x(1:50));
xlabel('t');
ylabel('x(t)');
title('The figure of the first fifty samples of x(t) versus the sampling times');
grid on;
[X,w]=ctfts(x,T);
figure;
plot(w,abs(X));
xlabel('w');
ylabel('The amplitude');
title('The figure of the magnitude of the continuous-time Fourier transform of the resonstructed signal');
grid on;
```

output:







($\Omega = 2\pi(2000)$)

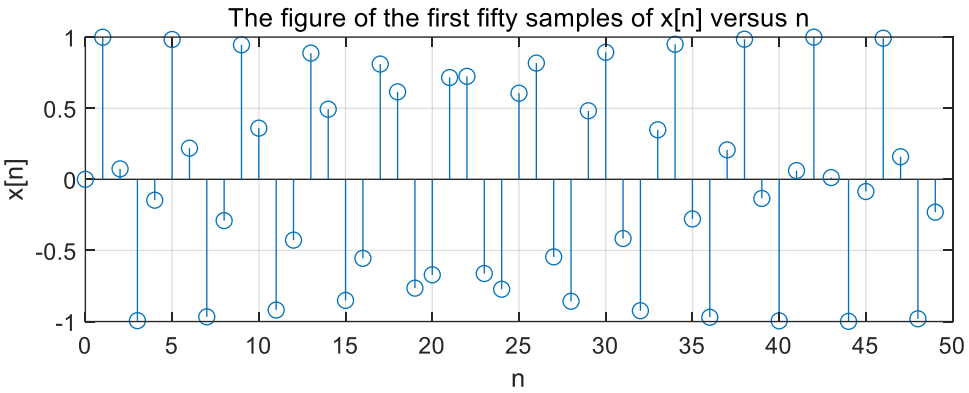
Code:

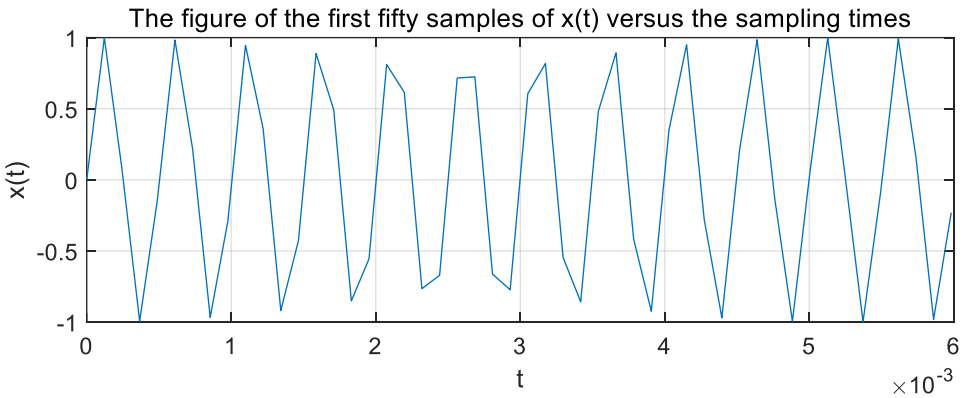
```
clc
clear all
n=[0:8191];
T=1/8192;
t=n./8192;
```

```
x=sin(4000*pi*t);
figure;
subplot(2,1,1);
stem(n(1:50),x(1:50));
xlabel('n');
ylabel('x[n]');
title('The figure of the first fifty samples of x[n] versus n');
grid on;
subplot(2,1,2);
plot(t(1:50),x(1:50));
xlabel('t');
ylabel('x(t)');
title('The figure of the first fifty samples of x(t) versus the sampling times');
grid on;
[X,w]=ctfts(x,T);
figure;
plot(w,abs(X));
xlabel('w');
ylabel('The amplitude');
title('The figure of the magnitude of the continuous-time Fourier transform of the resonstructed signal');
```

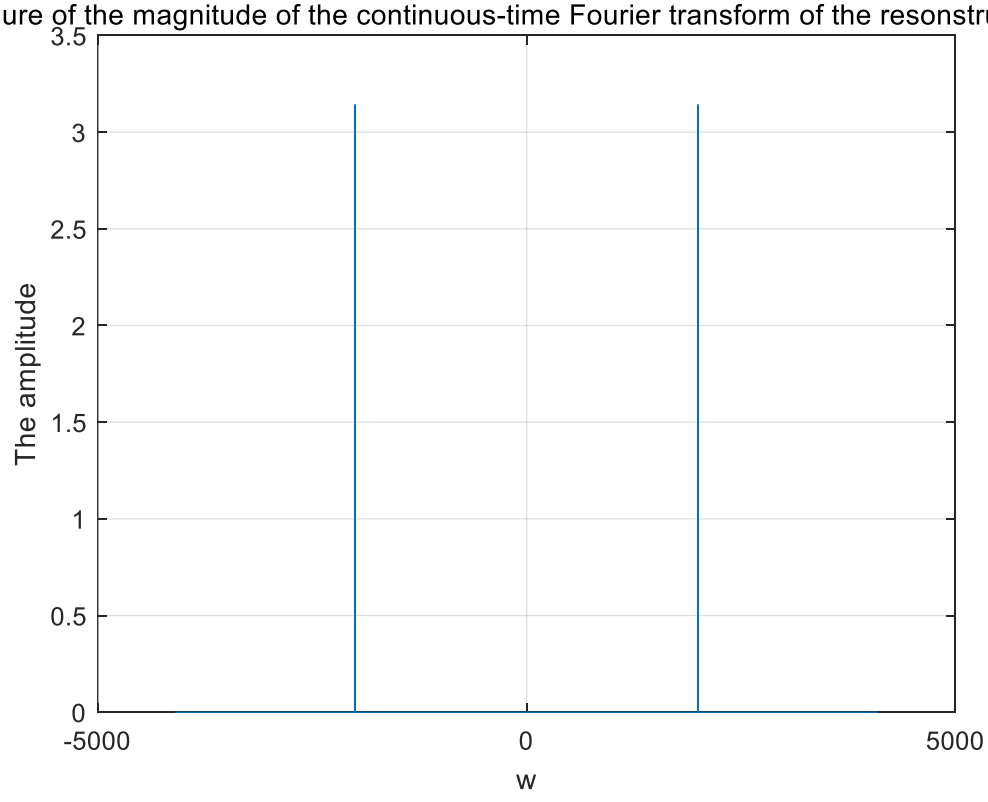
grid on;

output:





e figure of the magnitude of the continuous-time Fourier transform of the resonstructed



Answer:

The magnitude of X is nonzero at the expected frequencies ±1500 rad/s, confirming the signal's frequency content.

The phase of X is correct at these frequencies. At frequencies where the magnitude is negligible, the phase can be assumed zero due to numerical round-off errors.

2. Perform Single-Sideband AM with Hilbert Transform.

8.1 (c)(d)(e) (i)-(n)

c)answer:

The impulse response $h_\alpha[n]$ is a shifted version of $h[n]$ given by: $h_\alpha[n]=h[n-\alpha]$.

If α is an integer, the impulse response $h_\alpha[n]$ is symmetric about $n=\alpha$

de)

code:

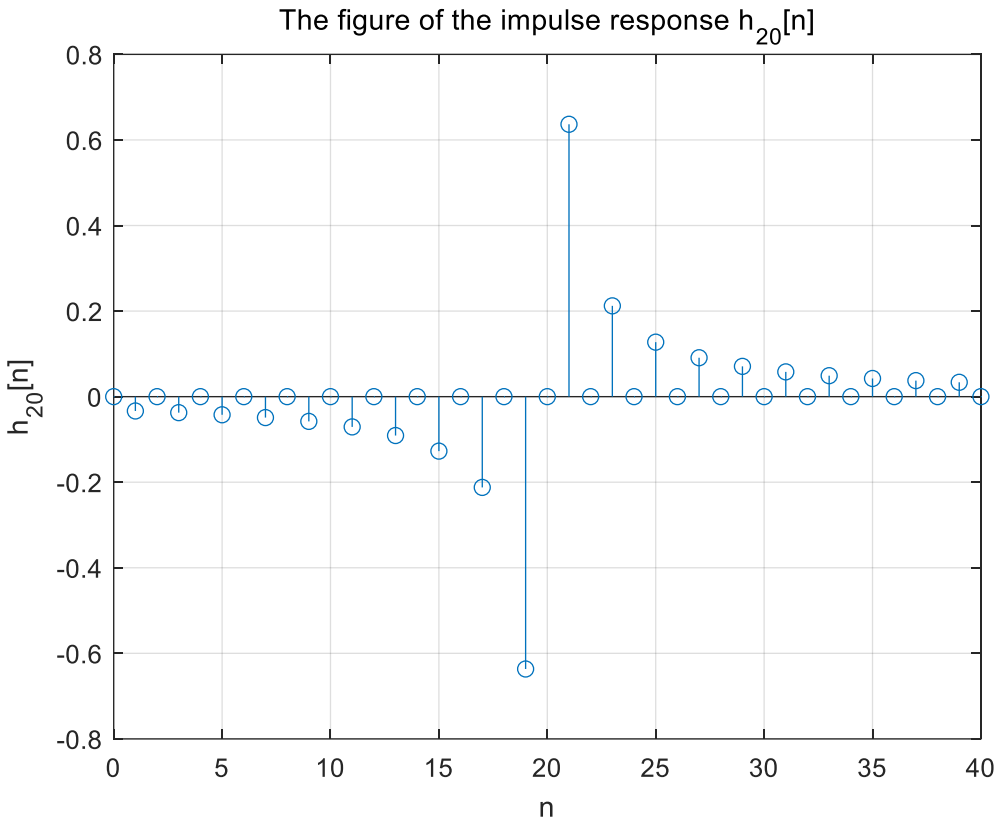
```
clc
clear all
n=[0:40];
h=[];
syms w
for i=0:1:40
h(i+1)=(1/(2*pi))*int(j*exp(-j*20*w+j*w*i),w,-
```

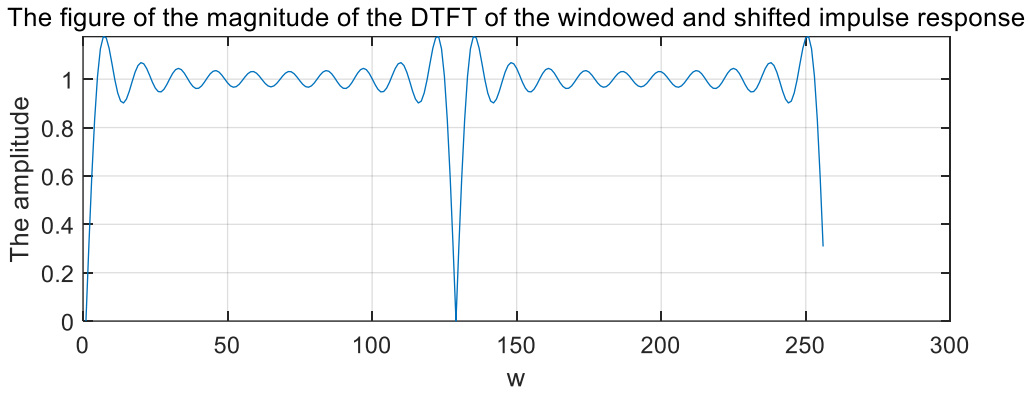
```
pi,0)+(1/(2*pi))*int(-j*exp(-j*20*w+j*w*i),w,0,pi);
end
figure;
stem(n,h);
xlabel('n');
ylabel('h_2_0[n]');
title('The figure of the impulse response h_2_0[n]');
grid on;
%.......(e).....%
figure;
subplot(2,1,1);
plot(abs(fft(h,256)));
xlabel('w');
ylabel('The amplitude');
title('The figure of the magnitude of the DTFT of the windowed
and shifted impulse response');
grid on;
subplot(2,1,2);
plot(unwrap(fftshift(angle(fft(h,256)))));
xlabel('w');
ylabel('The phase');
title('The figure of the phase of the DTFT of the windowed and
```

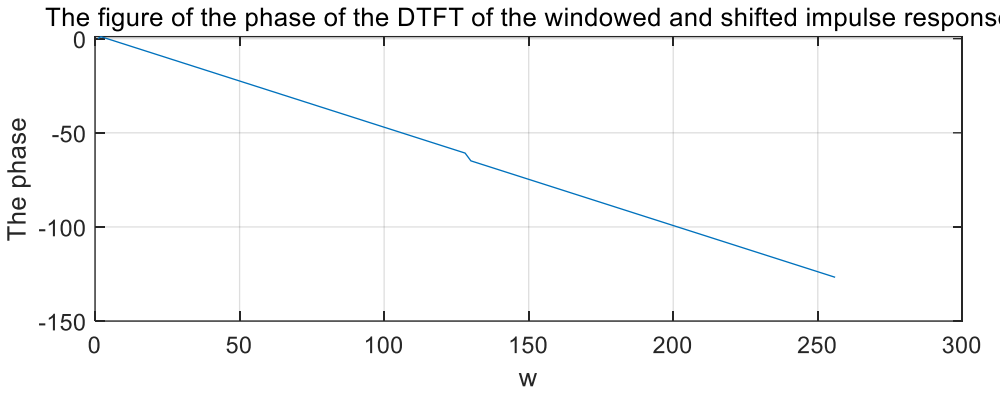
shifted impulse response');

grid on;

output:







Answer:

Magnitude: Matches the ideal Hilbert transform except for minor oscillations due to windowing.

Phase: The phase response is linearly decreasing due to the time shift of the impulse response, which aligns with the expected behavior.

Overall, the magnitude and phase of the DTFT of the windowed and shifted impulse response are consistent with the ideal Hilbert transform, except for slight deviations caused by windowing effects.

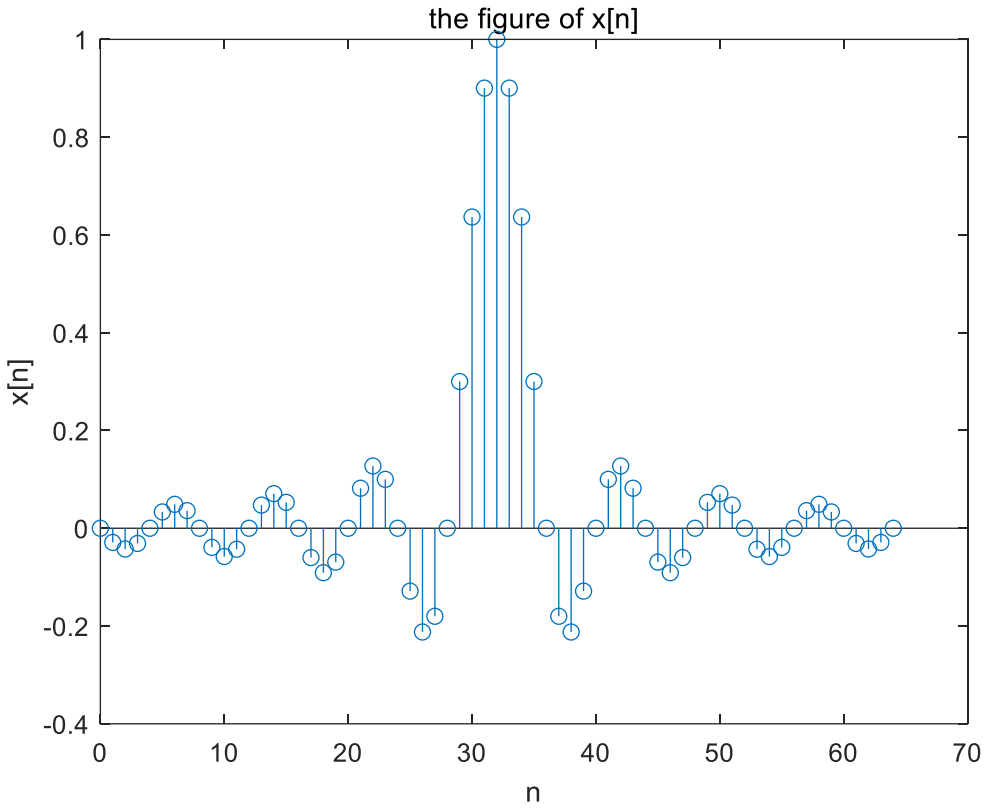
i)

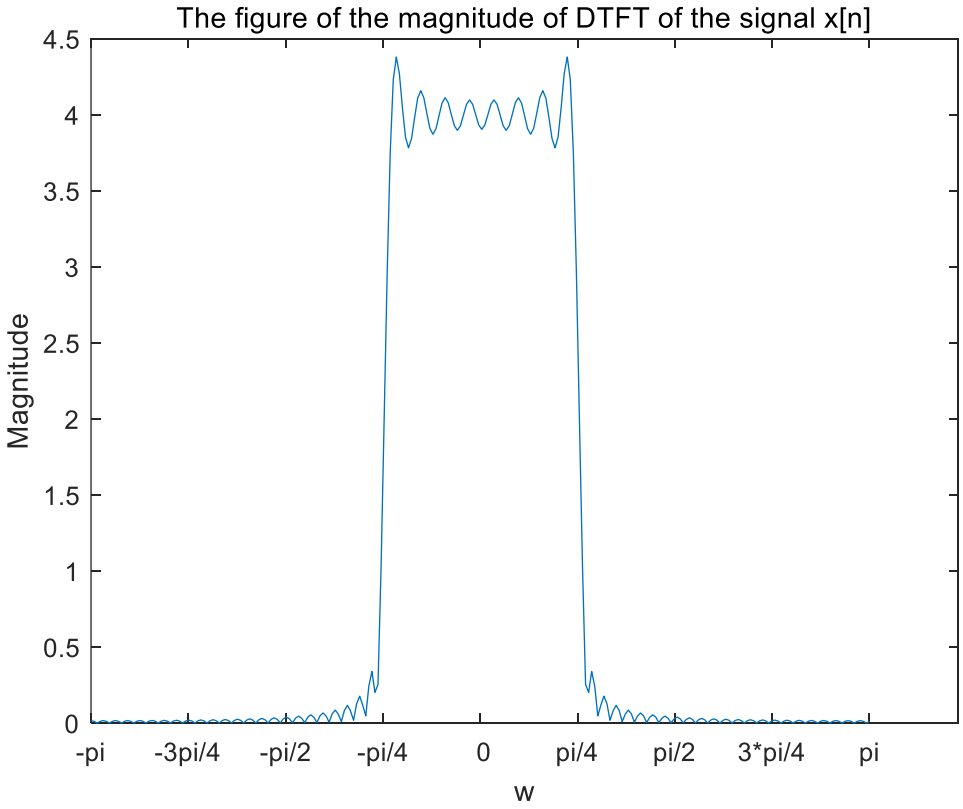
code:

```
clc
clear all
n=0:64;
x=(sin((pi/4).*(n-32))./((pi/4).*(n-32));
x(33)=1;
```

```
figure;
stem(n,x);
xlabel('n');
ylabel('x[n]');
title('the figure of x[n]');
X=fftshift(fft(x,256));
w = linspace(0,2*pi,256);
mag=abs(X);
figure;
plot(w,mag);
set(gca,'Xtick',[0,pi/4,pi/2,3*pi/4,pi,5*pi/4,3*pi/2,7*pi/4,2*pi]);
set(gca,'XTickLabel',{'-pi','-3pi/4','-pi/2','-pi/4','
0','pi/4','pi/2','3*pi/4','pi'});
xlabel('w');
ylabel('Magnitude');
title('The figure of the magnitude of DTFT of the signal x[n]');
```

output:





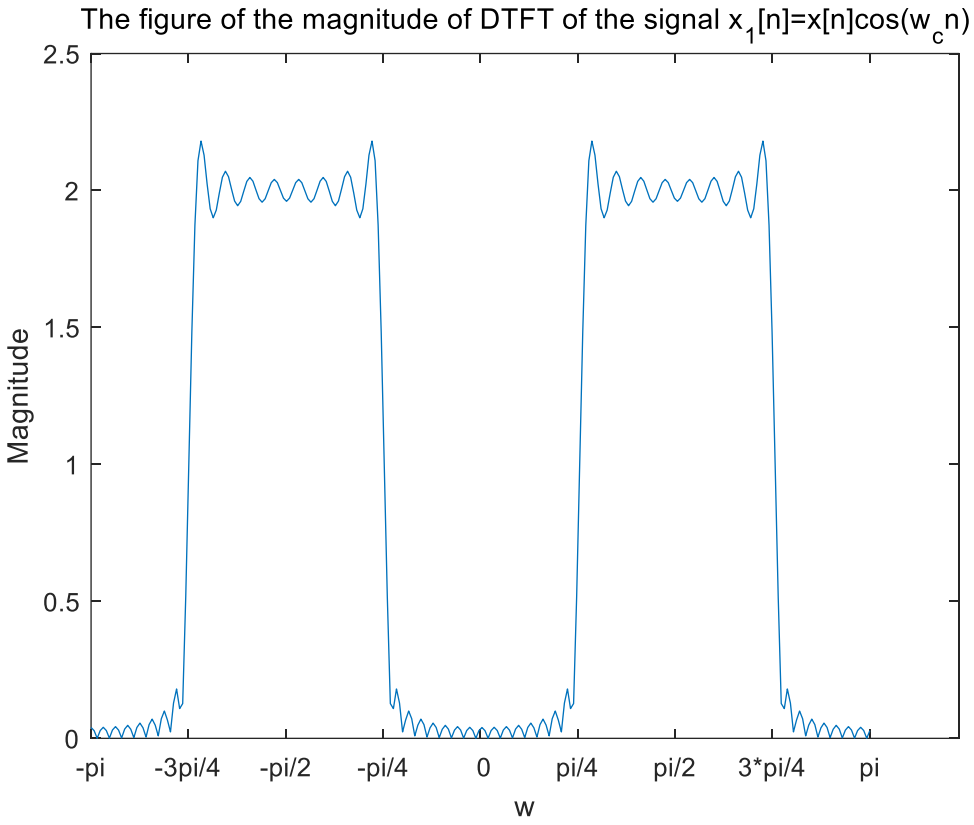
Answer:

The approximate frequency bandwidth of the signal x[n] is $\frac{\pi}{2}$

j)code:

```
clc
clear all
n=0:64;
x=(sin((pi/4).*(n-32))./((pi/4).*(n-32));
x(33)=1;
x1=cos((pi/2).*n).*x;
X1=fftshift(fft(x1,256));
mag1=abs(X1);
w = linspace(0,2*pi,256);
figure;
plot(w,mag1);
set(gca,'Xtick',[0,pi/4,pi/2,3*pi/4,pi,5*pi/4,3*pi/2,7*pi/4,2*pi]);
set(gca,'XTickLabel',{'-pi','-3pi/4','-pi/2','-pi/4','
0','pi/4','pi/2','3*pi/4','pi'});
xlabel('w');
ylabel('Magnitude');
title('The figure of the magnitude of DTFT of the signal
x_1[n]=x[n]cos(w_cn)');
```

output:



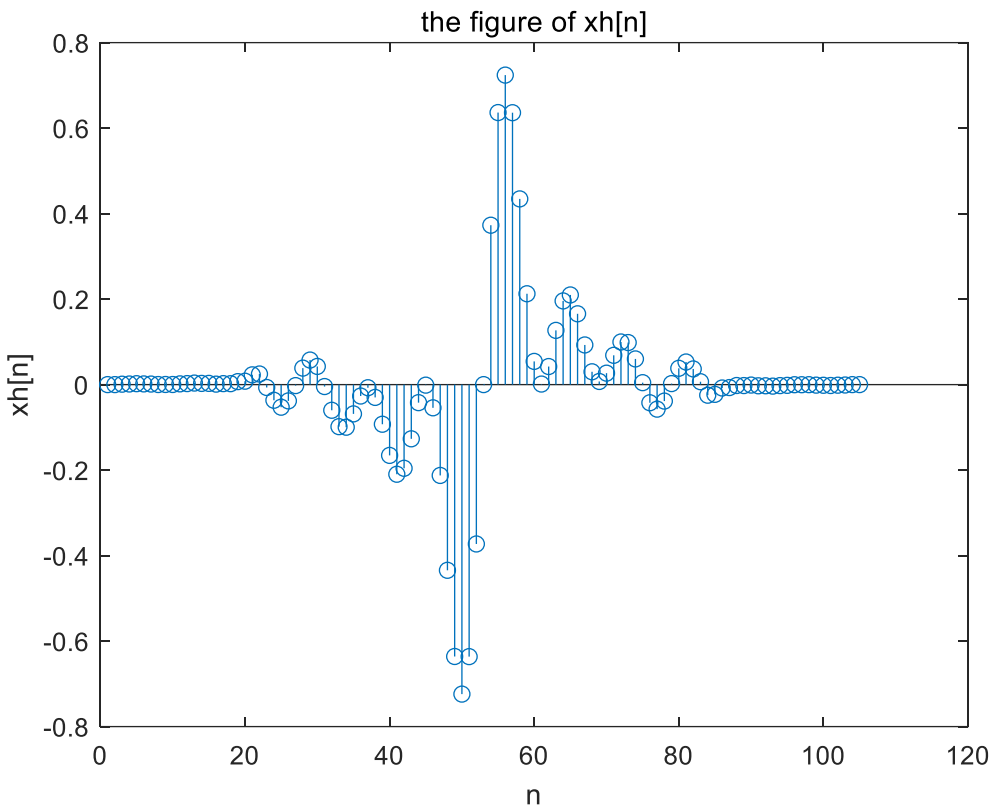
Kl)

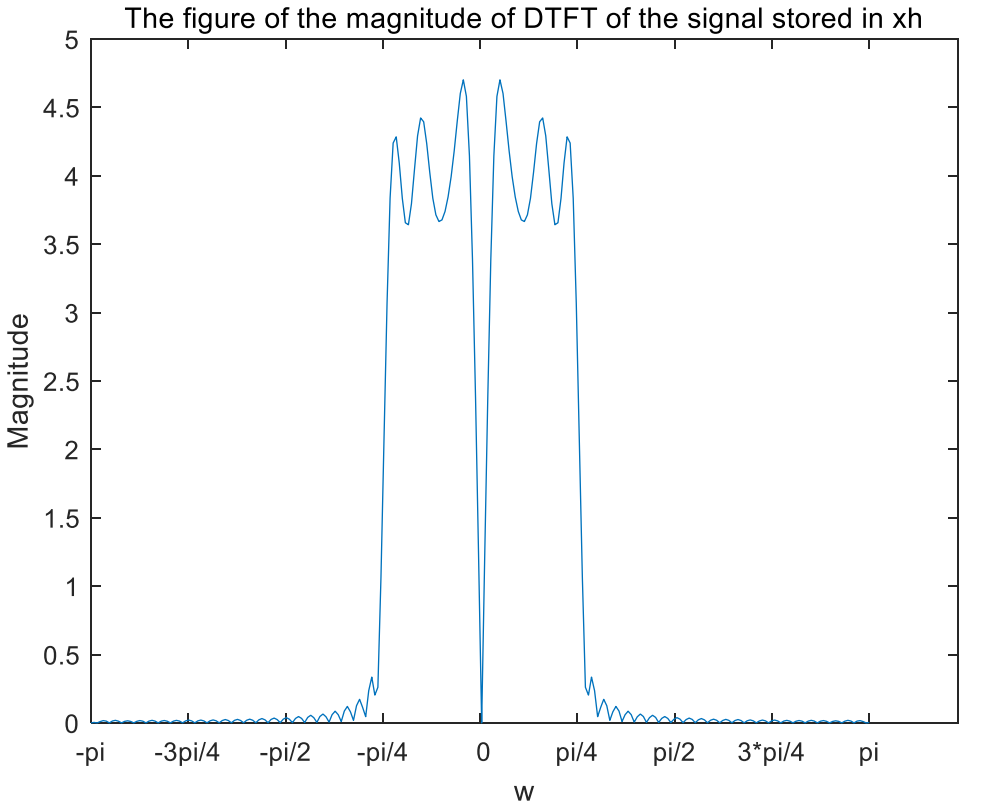
Code:

```
clc
clear all
n = 0:40;
h=[];
syms k
for i=0:1:40
h(i+1)=(1/(2*pi))*int(j*exp(-j*20*k+j*k*i),k,-
pi,0)+(1/(2*pi))*int(-j*exp(-j*20*k+j*k*i),k,0,pi);
end
w = linspace(0,2*pi,256);
```

```
n1=0:64;
x=(sin((pi/4).*(n1-32))./((pi/4).*(n1-32));
x(33)=1;
xh=conv(h,x);
figure;
stem(xh);
xlabel('n');
ylabel('xh[n]');
title('the figure of xh[n]');
XH=fftshift(fft(xh,256));
mag2=abs(XH);
figure;
plot(w,mag2);
set(gca,'Xtick',[0,pi/4,pi/2,3*pi/4,pi,5*pi/4,3*pi/2,7*pi/4,2*pi]);
set(gca,'XTickLabel',{'-pi','-3pi/4','-pi/2','-pi/4','
0','pi/4','pi/2','3*pi/4','pi'});
xlabel('w');
ylabel('Magnitude');
title('The figure of the magnitude of DTFT of the signal stored in
xh');
```

output:


the figure of xh[n]
xh[n]
n
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
0
20
40
60
80
100
120


The figure of the magnitude of DTFT of the signal stored in xh
Magnitude
w
5
4.5
4
3.5
3
2.5
2
1.5
1
0.5
0
-pi
-3pi/4
-pi/2
-pi/4
0
pi/4
pi/2
3*pi/4
pi

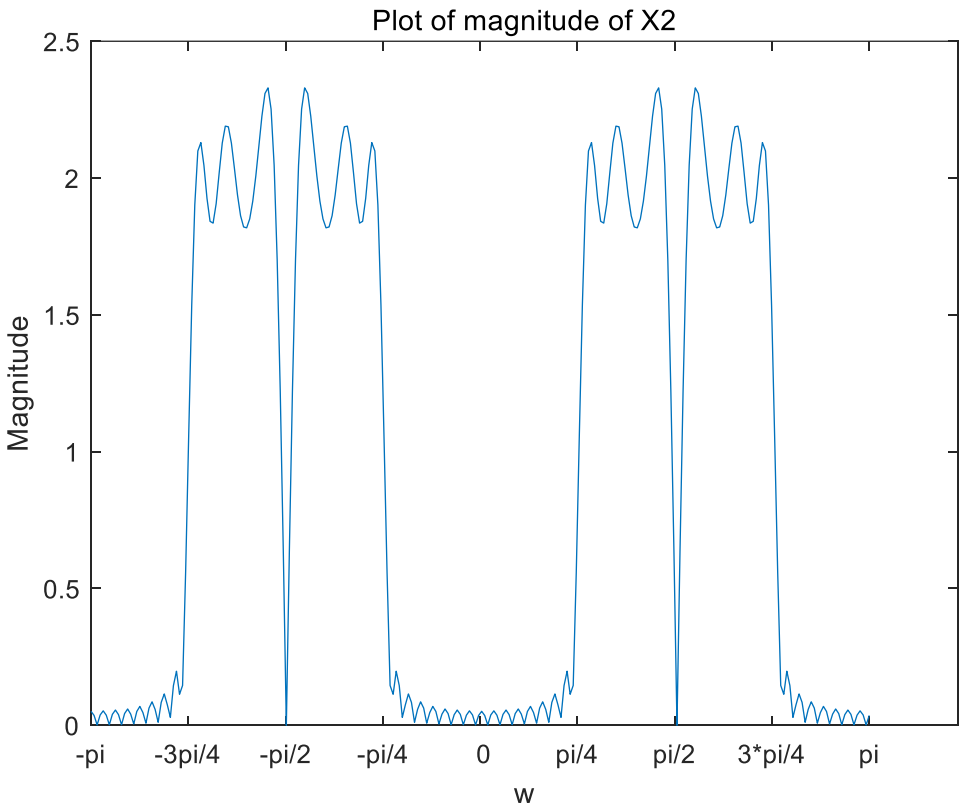
m)

code:

```
clc
clear all
n=0:40;
h=[];
syms k
for i=0:1:40
h(i+1)=(1/(2*pi))*int(j*exp(-j*20*k+j*k*i),k,-
pi,0)+(1/(2*pi))*int(-j*exp(-j*20*k+j*k*i),k,0,pi);
end
w = linspace(0,2*pi,256);
n1=0:64;
x=(sin((pi/4).*(n1-32))./((pi/4).*(n1-32));
x(33)=1;
xh=conv(h,x);
xh=xh(21:85);
x2=sin((pi/2).*n1).*xh;
X2=fftshift(fft(x2,256));
mag3=abs(X2);
figure;
plot(w,mag3);
```

```
set(gca,'Xtick',[0,pi/4,pi/2,3*pi/4,pi,5*pi/4,3*pi/2,7*pi/4,2*pi]);
set(gca,'XTickLabel',{'-pi','-3pi/4','-pi/2','-pi/4','
0','pi/4','pi/2','3*pi/4','pi'});
xlabel('w');
ylabel('Magnitude');
title('Plot of magnitude of X2');
```

output:



Answer:

The DTFT of x2[n] is the modulated version of x[n], with its frequency content shifted to $\pm\pi/2$

The bandwidth of x2[n] is the same as x[n], approximately $\pi/2$ radians

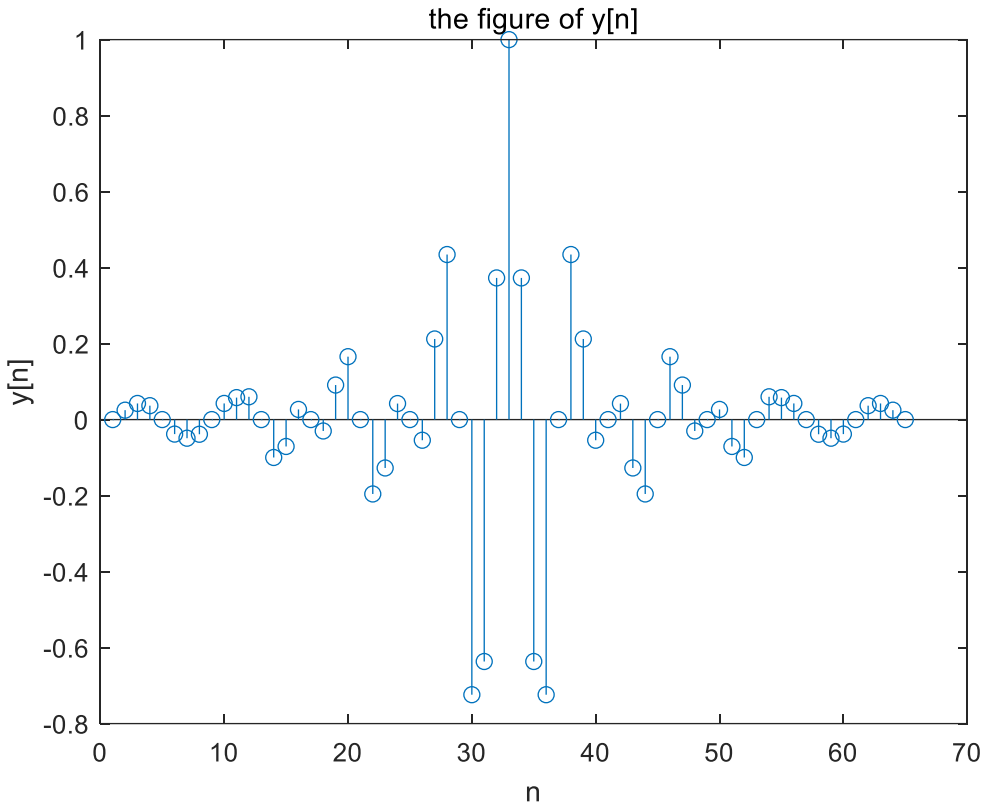
n)

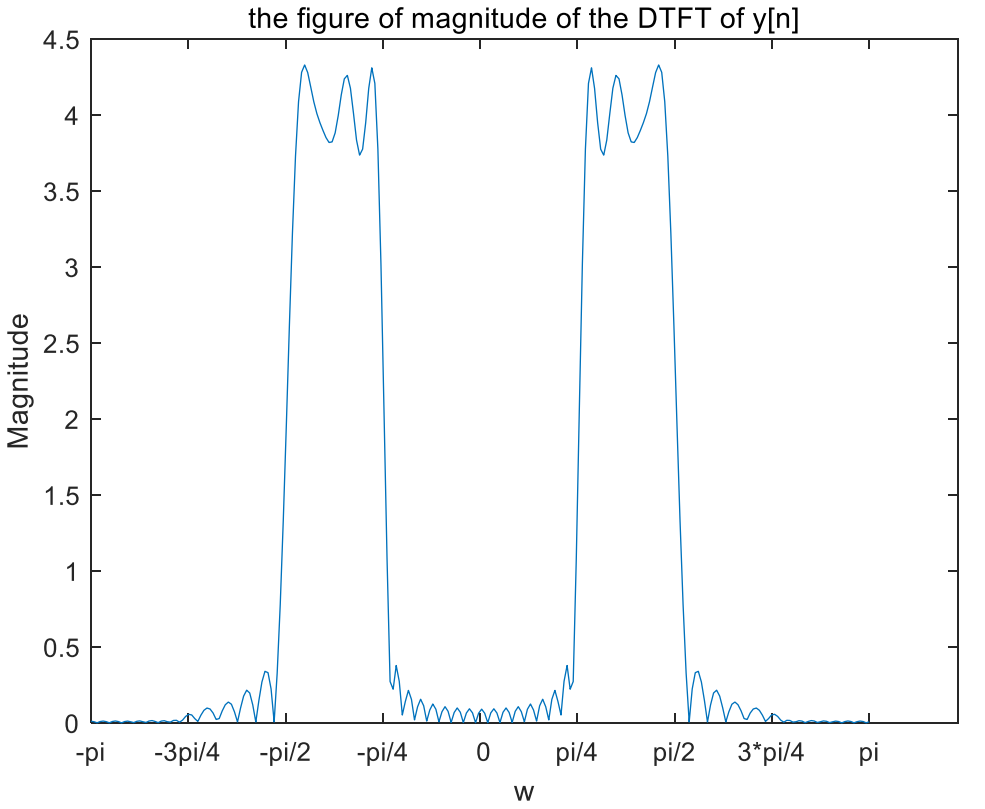
code:

```
clc
clear all
n=0:40;
h=[];
syms k
for i=0:1:40
h(i+1)=(1/(2*pi))*int(j*exp(-j*20*k+j*k*i),k,-
pi,0)+(1/(2*pi))*int(-j*exp(-j*20*k+j*k*i),k,0,pi);
end
w=linspace(0,2*pi,256);
n1=0:64;
x=(sin((pi/4).*(n1-32))./((pi/4).*(n1-32));
x(33)=1;
x1=cos((pi/2).*n1).*x;
xh=conv(h,x);
xh=xh(21:85);
x2=sin((pi/2).*n1).*xh;
y=x1+x2;
figure;
stem(y);
```

```
xlabel('n');
ylabel('y[n]');
title('the figure of y[n]');
Y=fftshift(fft(y,256));
mag4=abs(Y);
figure;
plot(w,mag4);
set(gca,'Xtick',[0,pi/4,pi/2,3*pi/4,pi,5*pi/4,3*pi/2,7*pi/4,2*pi]);
set(gca,'XTickLabel',{'-pi','-3pi/4','-pi/2','-pi/4','
0','pi/4','pi/2','3*pi/4','pi'});
xlabel('w');ylabel('Magnitude');title('the figure of magnitude of
the DTFT of y[n]');
```

output:





Answer:

Where does the DTFT of y[n] approach zero?

- From the DTFT magnitude plot, the DTFT of y[n] approaches zero for frequencies $|\omega| > \pi/2$
- This indicates that the frequency content of y[n]] is primarily concentrated within the range $-\pi/2$ to $\pi/2$.

Comparison of the bandwidth of y[n] and the original signal x[n], verifying SSB modulation:

- The original signal x[n] has a bandwidth of approximately $\pi/2$ radians/sample.
- The DTFT of y[n]y[n]y[n] retains the same bandwidth but shifts the frequency content to around $-\pi/4$ and $\pi/4$ preserving only one sideband.
- This confirms the characteristic of single-sideband (SSB) modulation, where one sideband is translated to a higher frequency while maintaining the same bandwidth.

3. Make pole-zero plot using MATLAB.

9.1 (a)(c) 10.1(a)(b)

9.1a)

```
clc
clear all
%.....(i).....%
b1=[1 5];
a1=[1 2 3];
zs1=roots(b1)
ps1=roots(a1)
figure;
plot(real(zs1),imag(zs1),'o');
xlabel('The real part');
ylabel('The imaginary part');
title('The zero-pole diagram of (i)');
hold on;
plot(real(ps1),imag(ps1),'x');
xlabel('The real part');
ylabel('The imaginary part');
title('The zero-pole diagram of (i)');
axis([-6 6 -6 6]);
grid on;
```

```
%.....(ii).....%
b2=[2 5 12];
a2=[1 2 10];
zs2=roots(b2)
ps2=roots(a2)
figure;
plot(real(zs2),imag(zs2),'o');
xlabel('The real part');
ylabel('The imaginary part');
title('The zero-pole diagram of (ii)');
hold on;
plot(real(ps2),imag(ps2),'x');
xlabel('The real part');
ylabel('The imaginary part');
title('The zero-pole diagram of (ii)');
axis([-6 6 -6 6]);
grid on;
%.....(iii).....%
b3=[2 5 12];
a3=[1 4 14 20];
zs3=roots(b3)
ps3=roots(a3)
```

```
figure;
plot(real(zs3),imag(zs3),'o');
xlabel('The real part');
ylabel('The imaginary part');
title('The zero-pole diagram of (iii)');
hold on;
plot(real(ps3),imag(ps3),'x');
xlabel('The real part');
ylabel('The imaginary part');
title('The zero-pole diagram of (iii)');
axis([-6 6 -6 6]);
grid on;
```

output

```
zs1 =

     -5


ps1 =

  -1.0000 + 1.4142i
  -1.0000 - 1.4142i
```

zs2 =

-1.2500 + 2.1065i

-1.2500 - 2.1065i

ps2 =

-1.0000 + 3.0000i

-1.0000 - 3.0000i

zs3 =

-1.2500 + 2.1065i

-1.2500 - 2.1065i

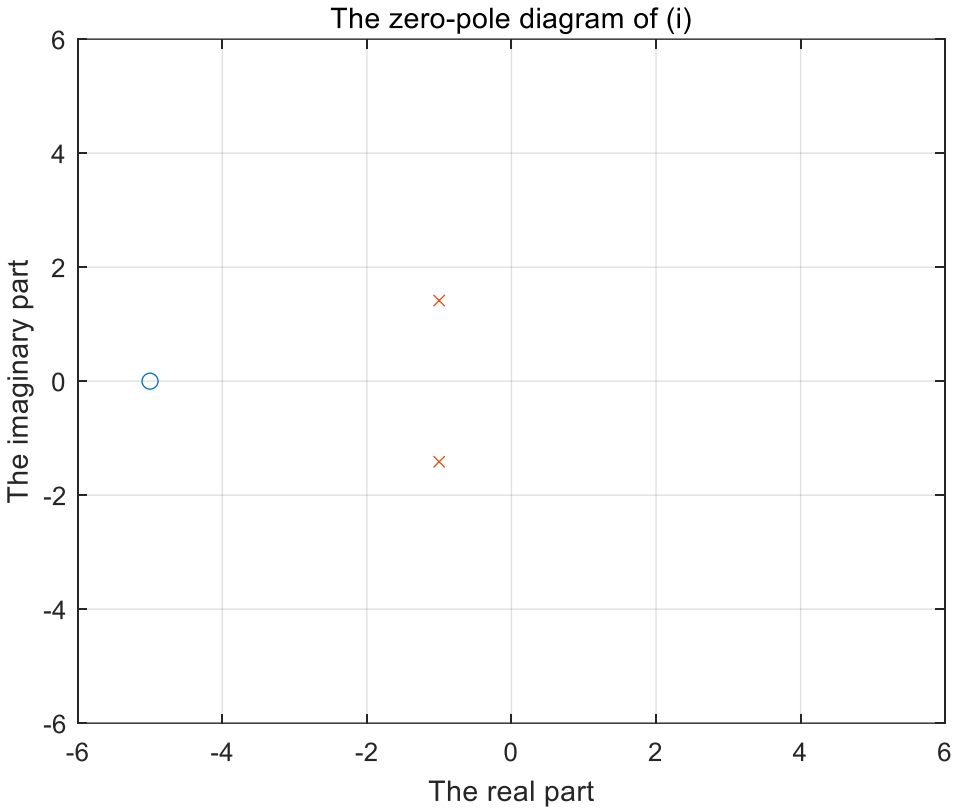
ps3 =

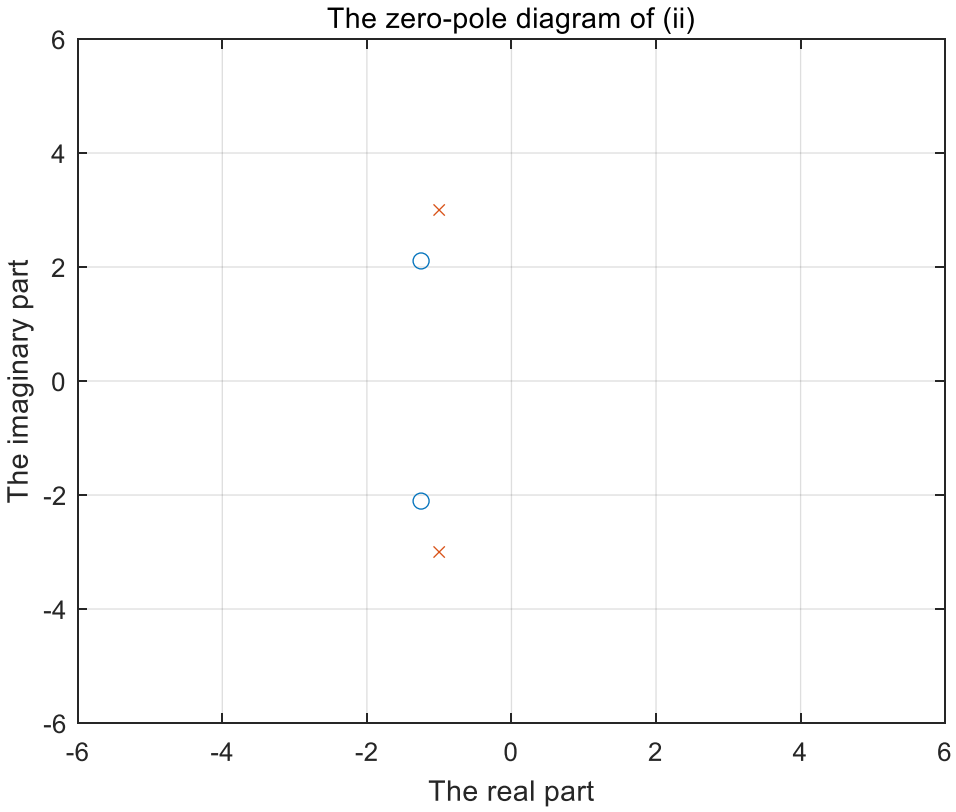
-1.0000 + 3.0000i

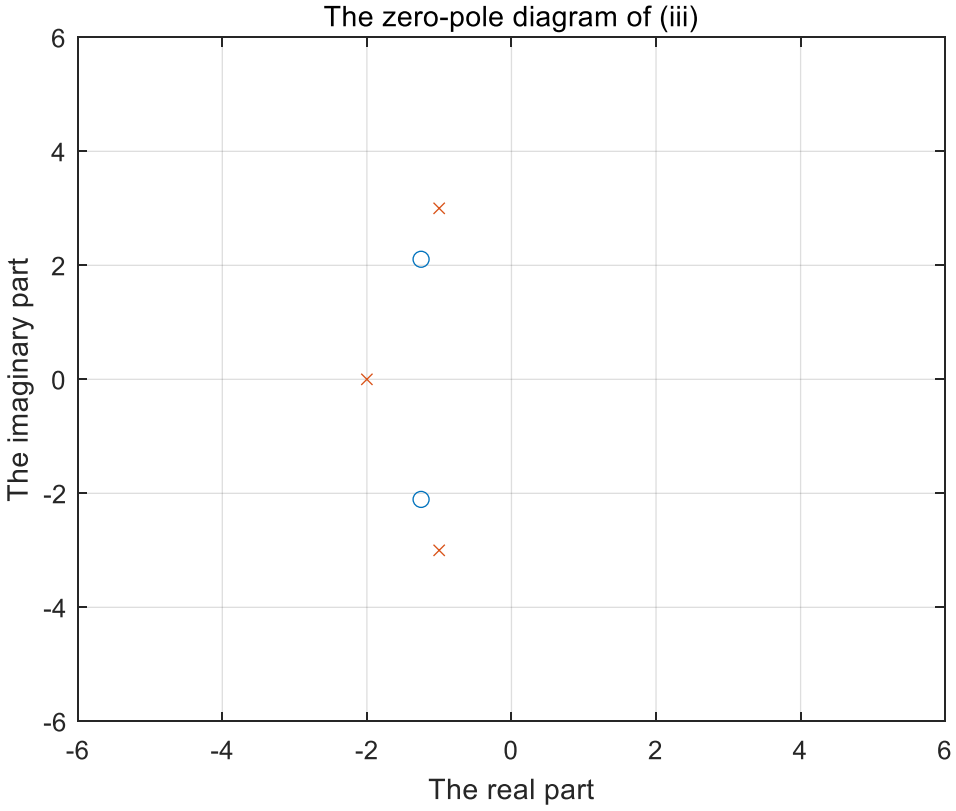
-1.0000 - 3.0000i

-2.0000 + 0.0000i

>>




The zero-pole diagram of (ii)
The imaginary part
The real part
6
4
2
0
-2
-4
-6
-6
-4
-2
0
2
4
6


The zero-pole diagram of (iii)
The imaginary part
The real part
6
4
2
0
-2
-4
-6
-6
-4
-2
0
2
4
6

C)

Code:

```
clc
clear all
a=[1 -3];
b=[1 2 5];
zs=roots(b)
ps=roots(a)
figure;
plot(real(zs),imag(zs),'o');
xlabel('The real part');
ylabel('The imaginary part');
title('The zero-pole diagram');
hold on;
plot(real(ps),imag(ps),'x');
xlabel('The real part');
ylabel('The imaginary part');
title('The zero-pole diagram');
axis([-4 4 -4 4]);
grid on;
```

output:

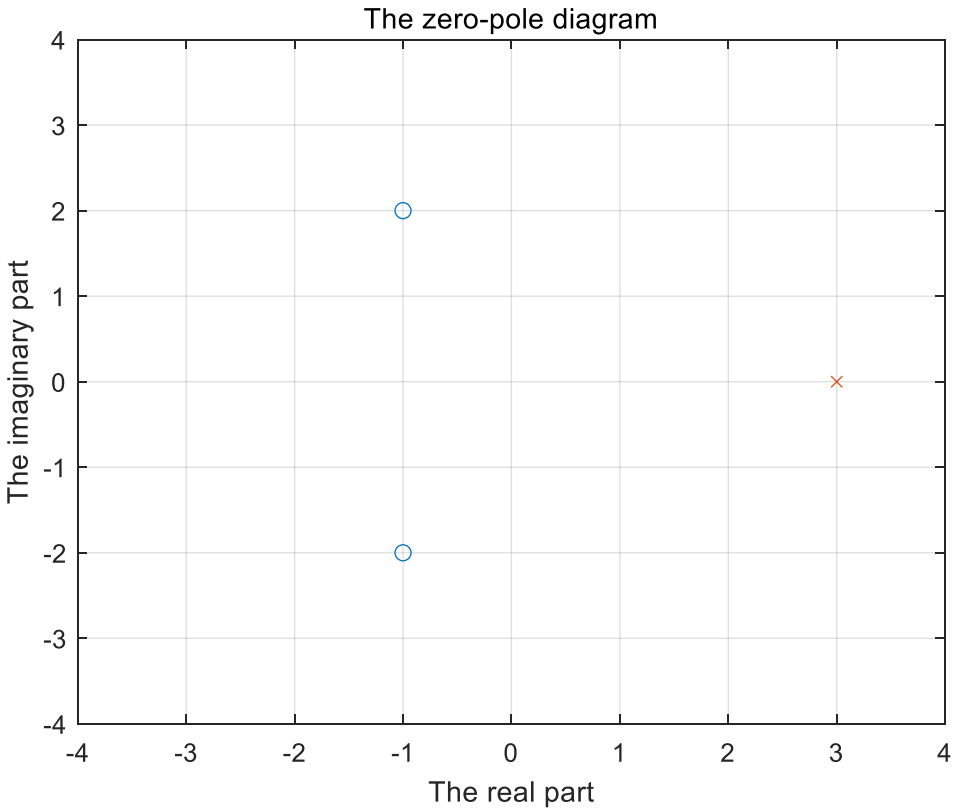
zs =

-1.0000 + 2.0000i

-1.0000 - 2.0000i

ps =

3



10.1

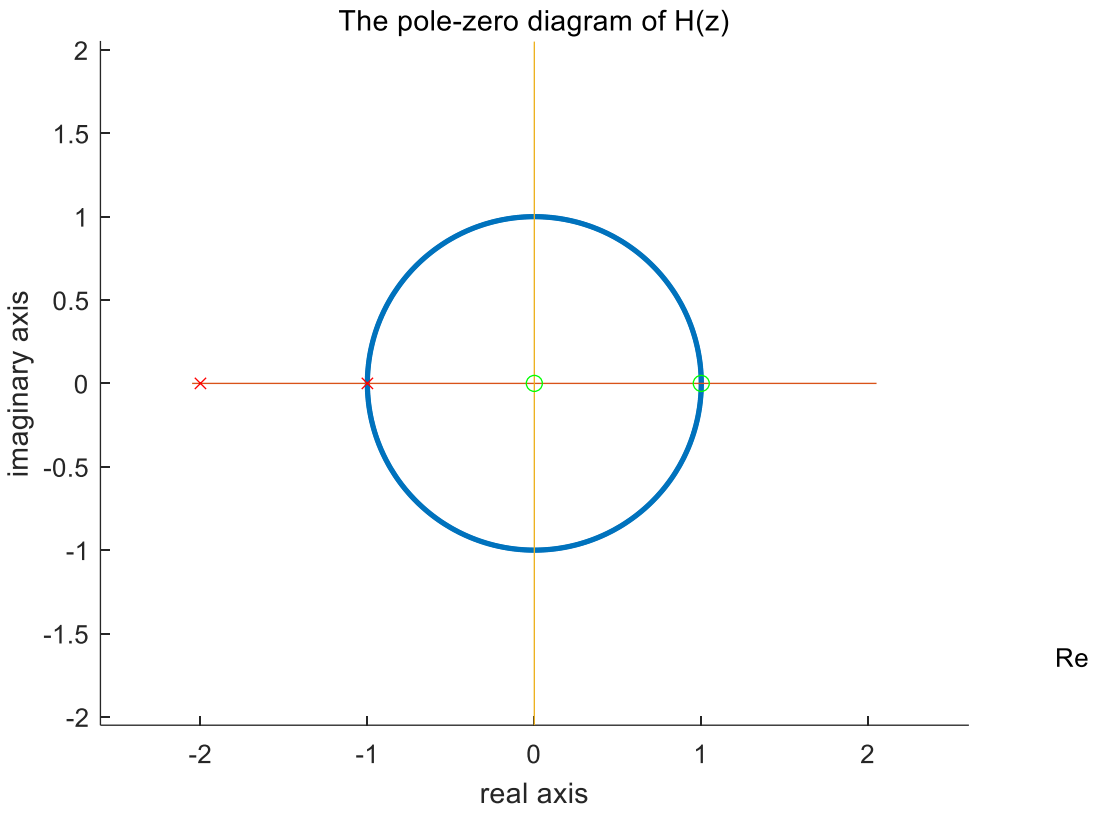
a)

code:

```
b1=[1 -1 0];
a1=[1 3 2];
figure;
dpzplot(b1,a1);
xlabel('real axis');
ylabel('imaginary axis');
title('The pole-zero diagram of H(z)');
```

output:

....



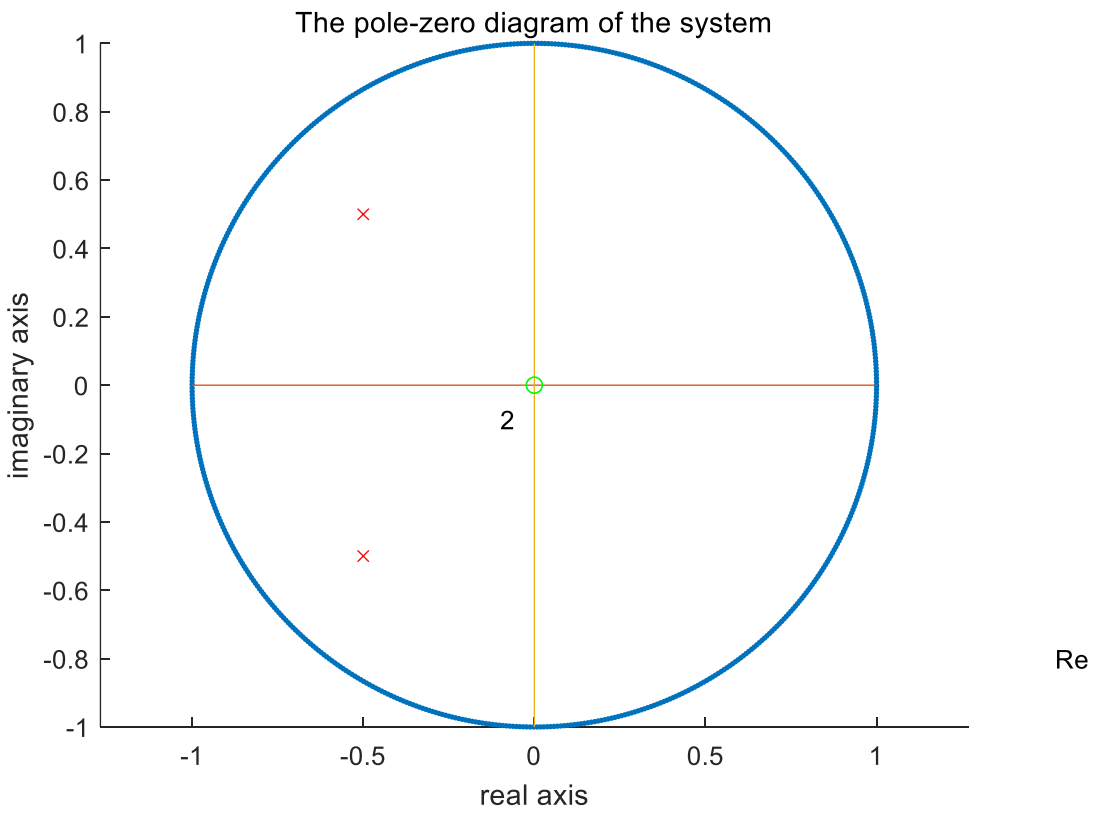
b)

code:

```
b2=[1 0 0];
a2=[1 1 0.5];
figure;
dpzplot(b2,a2);
xlabel('real axis');
ylabel('imaginary axis');
title('The pole-zero diagram of the system');
```

output:

....



4. Obtain the frequency response of a second-order system.

9.2 (a)(b)

a)

Code:

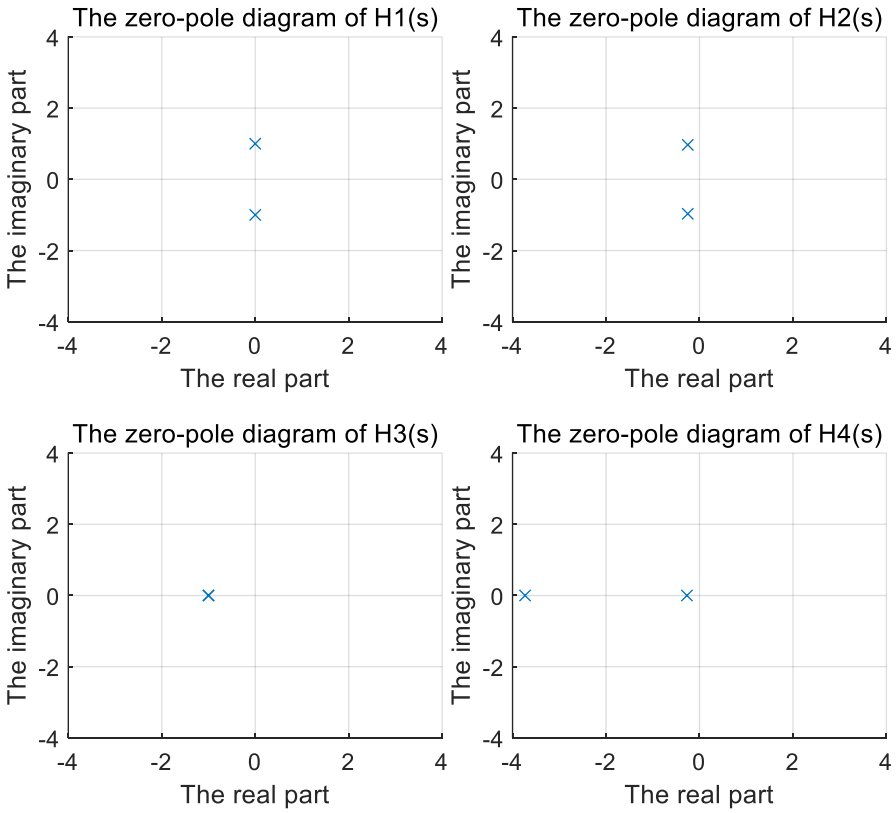
```
clc
clear all
b=1;
a1=[1 0 1];
zs=roots(b);
ps1=roots(a1);
subplot(2,2,1);
plot(real(zs),imag(zs),'o');
xlabel('The real part');
ylabel('The imaginary part');
title('The zero-pole diagram of H1(s)');
hold on;
plot(real(ps1),imag(ps1),'x');
xlabel('The real part');
ylabel('The imaginary part');
title('The zero-pole diagram of H1(s)');
axis([-4 4 -4 4]);
grid on;
```

```
a2=[1 1/2 1];
zs=roots(b);
ps2=roots(a2);
subplot(2,2,2);
plot(real(zs),imag(zs),'o');
xlabel('The real part');
ylabel('The imaginary part');
title('The zero-pole diagram of H2(s)');
hold on;
plot(real(ps2),imag(ps2),'x');
xlabel('The real part');
ylabel('The imaginary part');
title('The zero-pole diagram of H2(s)');
axis([-4 4 -4 4]);
grid on;
a3=[1 2 1];
zs=roots(b);
ps3=roots(a3);
subplot(2,2,3);
plot(real(zs),imag(zs),'o');
xlabel('The real part');
ylabel('The imaginary part');
```

```
title('The zero-pole diagram of H3(s)');
hold on;
plot(real(ps3),imag(ps3),'x');
xlabel('The real part');
ylabel('The imaginary part');
title('The zero-pole diagram of H3(s)');
axis([-4 4 -4 4]);
grid on;
a4=[1 4 1];
zs=roots(b);
ps4=roots(a4);
subplot(2,2,4);
plot(real(zs),imag(zs),'o');
xlabel('The real part');
ylabel('The imaginary part');
title('The zero-pole diagram of H4(s)');
hold on;
plot(real(ps4),imag(ps4),'x');
xlabel('The real part');
ylabel('The imaginary part');
title('The zero-pole diagram of H4(s)');
axis([-4 4 -4 4]);
```

grid on;

output:



b)

code:

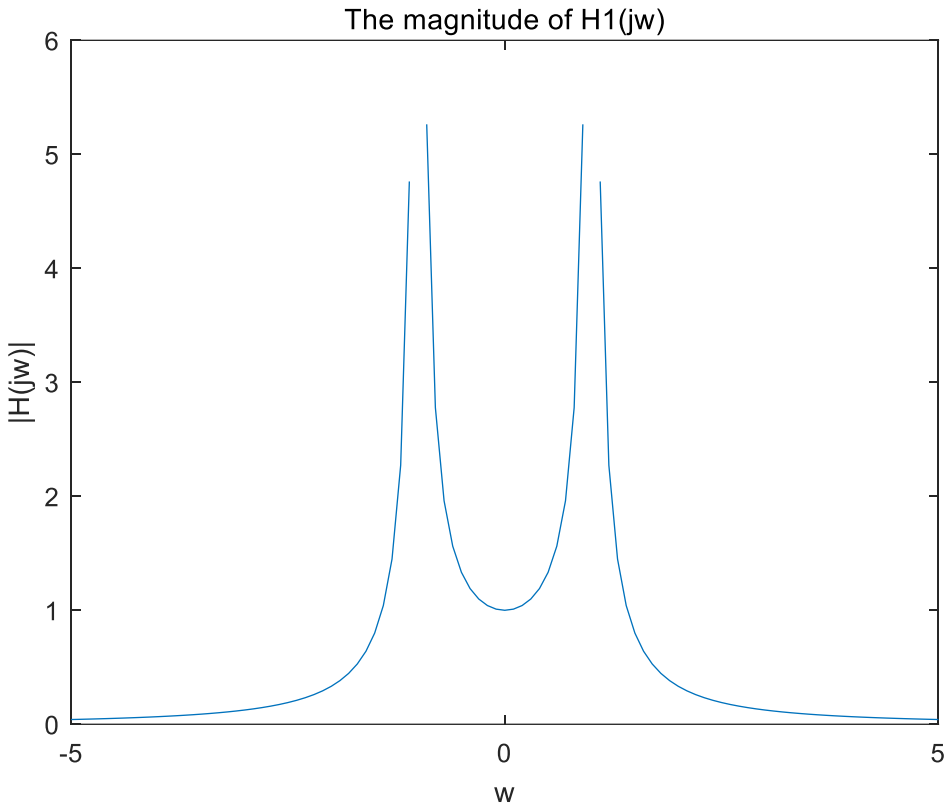
```
clc
clear all
w=-5:0.1:5;
b=1;
a1=[1 0 1];
a2=[1 1/2 1];
a3=[1 2 1];
a4=[1 4 1];
```

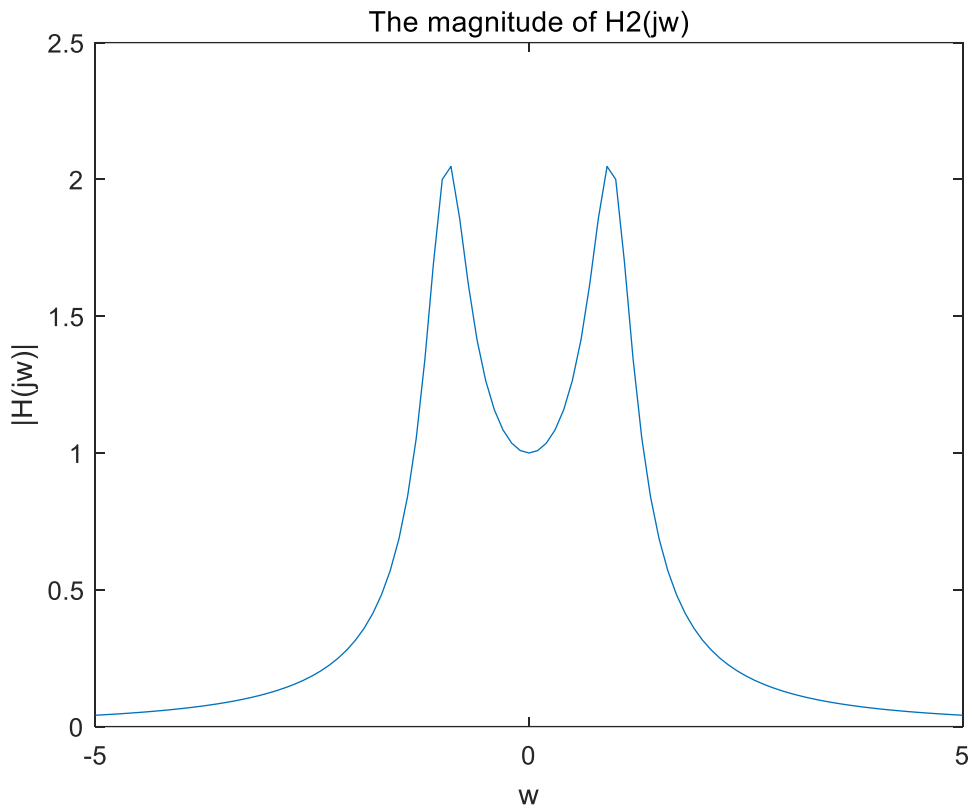
```
f1=freqs(b,a1,w);
f2=freqs(b,a2,w);
f3=freqs(b,a3,w);
f4=freqs(b,a4,w);
figure;
plot(w,abs(f1));
xlabel('w');
ylabel('|H(jw)|');
title('The magnitude of H1(jw)');
figure;
plot(w,abs(f2));
xlabel('w');
ylabel('|H(jw)|');
title('The magnitude of H2(jw)');
figure;
plot(w,abs(f3));
xlabel('w');
ylabel('|H(jw)|');
title('The magnitude of H3(jw)');
figure;
plot(w,abs(f4));
xlabel('w');
```

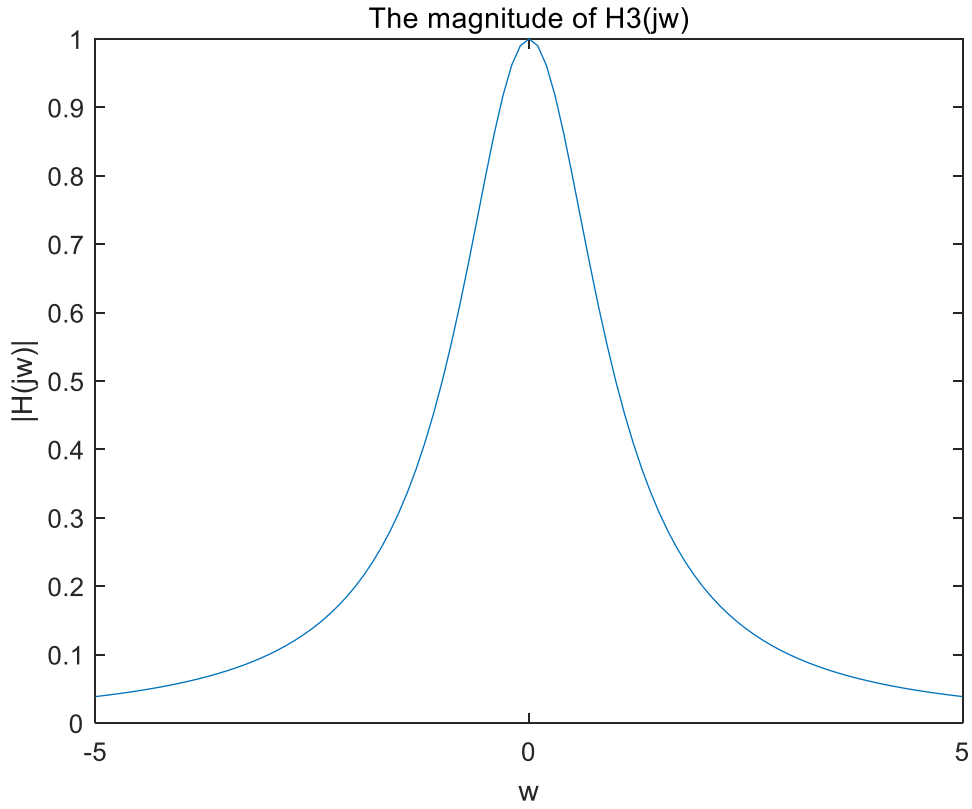
ylabel('|H(jw)|');

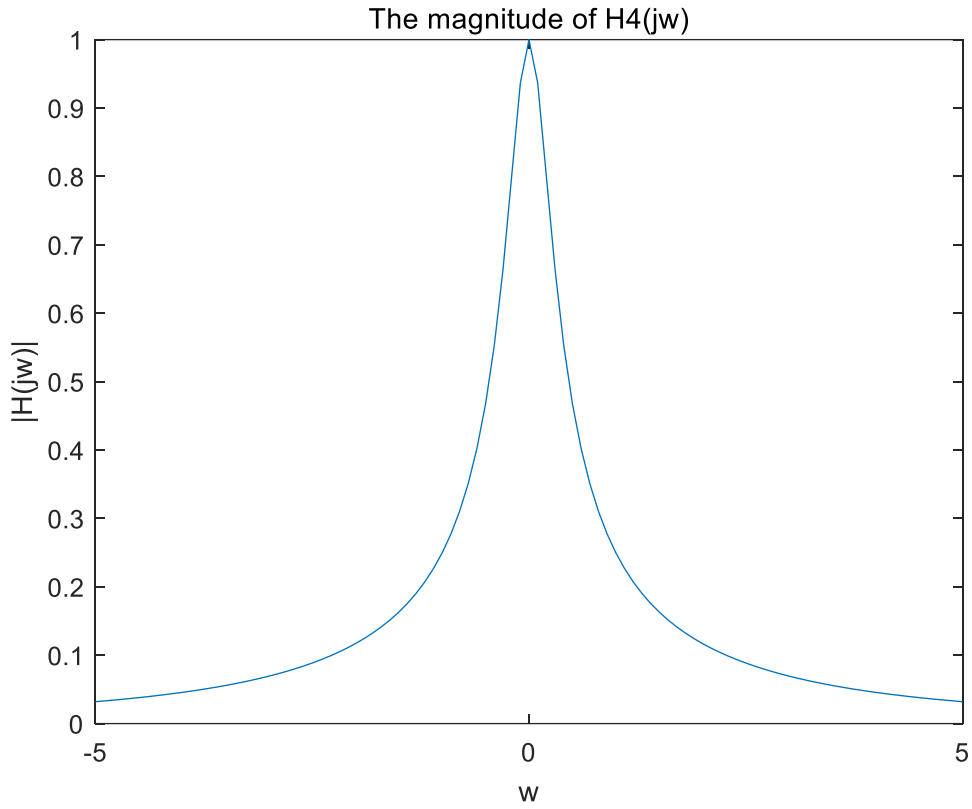
title('The magnitude of H4(jw)');

output:




The magnitude of H2(jw)
2.5
2
1.5
1
0.5
0
|H(jw)|
-5
0
5
w


The magnitude of H3(jw)
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
|H(jw)|
-5
0
5
w



Answer:

1. Pole Locations and Their Effects:

The pole-zero diagrams for H1(s), H2(s), H3(s), and H4(s) demonstrate the effect of the damping ratio ζ\zetaζ:

- H1(s) (ζ=0):
    - Poles are purely imaginary (on the jωj\omegajω-axis).
    - The system is undamped, resulting in resonant peaks in the frequency response.
- H2(s) (ζ=1/4):
    - Poles are complex conjugates in the left-half plane but close to the imaginary axis.
    - The system is underdamped, producing noticeable

resonant peaks with reduced amplitude compared to H1(s).

- H3(s) (ζ=1):
  - Poles are real and repeated on the negative real axis.
  - The system is critically damped, resulting in no resonance and a smooth peak in the frequency response.
- H4(s) (ζ=2:
  - Poles are real and distinct on the negative real axis.
  - The system is overdamped, leading to a further flattened frequency response with no resonant peaks.

2. Frequency Response Comparisons:

From the magnitude plots:

- H1(jω) (ζ=0):
  - Shows sharp resonant peaks at non-zero frequencies.
  - This occurs because the poles are on the jω axis, amplifying the frequency response at resonance.
- H2(jω) (ζ=1/4):
  - The resonant peaks are still present but less sharp due to the increased damping.
  - The poles move slightly into the left-half plane, reducing the peak amplitude.

- H3(jω) (ζ=1):
    - Resonance disappears, and the response has a smooth peak at ω=0.
    - This corresponds to the critically damped system, where the poles are real and repeated.
- H4(jω) (ζ=2):
    - The response is further flattened, with no resonance and a smooth decay.
    - The poles are real, distinct, and far into the left-half plane, minimizing frequency-domain amplification.

3. Why $H(j\omega)_{\omega=0}$ is the Same:
    - At ω=0 (DC), the transfer function H(s) simplifies to:
    - $H(j0) = \frac{{\omega_n}^2}{0+2\xi\omega_n(0)+{\omega_n}^2}$=1
    - Since the numerator and denominator are independent of ζat ω=0, the magnitude $|H(j\omega)|_{\omega=0}$ is the same for all systems.